

ADAMAS: an open, tested framework for diamond electronics and room-temperature nitrogen-vacancy quantum processors

Aleksander Norman

California State University, Dominguez Hills
github.com/Normansrule/adamas-diamond-framework

Preprint, generated from the repository

Abstract

Diamond combines the largest breakdown field and thermal conductivity of any practical semiconductor with a defect, the nitrogen-vacancy (NV) center, whose electron spin is a qubit at room temperature. ADAMAS is an open framework that follows one path from wafer to quantum processor with executable, cited models: 537 references (500 verified against Crossref), 48 regenerable figures, a process design kit with a placed 4-bit processor, circuit-level quantum error correction, and hands-on experiments from US\$100 upward. Three results stand out. (i) Measured hydrogen-terminated diamond transistors already beat silicon’s one-dimensional limit by $\sim 89\times$, and in an 800 V traction inverter an ideal diamond switch loses 3.8 W against 29.7 W for silicon carbide. (ii) Evaluated with the published parameters of the 2013 room-temperature entangled pair, coherence alone would permit a Bell fidelity of 0.998; the measured 0.67 corresponds to a charge-state preparation probability of 0.75, identifying preparation, not coherence, as the bottleneck. (iii) Circuit-level simulations with idle errors during millisecond readout show that surface-code scaling survives only while readout stays below about a tenth of the nuclear-memory coherence time, that a bias-tailored XZZX code extends this about tenfold, and that a factoring-scale machine fits on a 5.7 mm die but runs for years. Every number in this paper is generated from the package.

1 Introduction

Silicon’s dominance rests on process maturity, not material merit. For power conversion, high-temperature electronics, and room-temperature quantum technology, diamond’s figures of merit are extreme: its Baliga figure of merit for power switches is $48,976\times$ silicon’s and its thermal conductivity $14.7\times$ [1, 2]. Its obstacles are equally concrete: boron acceptors sit 0.37 eV deep, so under 1% ionize at room temperature [3]; single-crystal wafers only recently reached 3 inches [4]; and logic has been p-channel only until the first n-channel MOSFET in 2024 [5].

ADAMAS treats these facts as inputs to models that can be run, tested, and falsified. The contribution is not a new device but a connected, reproducible chain: materials $\rightarrow$ process $\rightarrow$ circuits $\rightarrow$ qubits $\rightarrow$ error correction $\rightarrow$ machine size $\rightarrow$ experiments, with each link checked against published data or an independent tool (ngspice, Icarus Verilog, Yosys, KLayout, Stim, PyMatching).

2 Power devices and converters

The unipolar limit $R_{on,sp} = 4BV^2/(\epsilon\mu E_c^3)$ [1] places ideal diamond far below every other material (Fig. 1). Measured lateral hole-gas transistors reach 3659 V [8] and, at 2568 V, $7.54\text{ m}\Omega\text{ cm}^2$ [7], about $89\times$ better than silicon’s limit. For an area-optimized hard-switched leg, conduction and switching losses equalize and the loss is $P = 2IV\sqrt{fR_{on,sp}C_{oss,sp}/2}$ [9]. Table 1 shows the result for an 800 V, 300 A, 20 kHz traction inverter; bulk-doped diamond additionally improves with temperature as boron ionizes [10].

Switch (1.2 kV class)	Loss per switch (W)
Silicon, ideal	209.5
Silicon carbide, ideal	29.7
Gallium nitride, ideal	20.6
Diamond, ideal	3.8
Diamond, 2022 device scaled	89

Table 1: Area-optimized switch loss in an 800 V, 300 A, 20 kHz inverter leg (first order: no gate charge or reverse recovery).

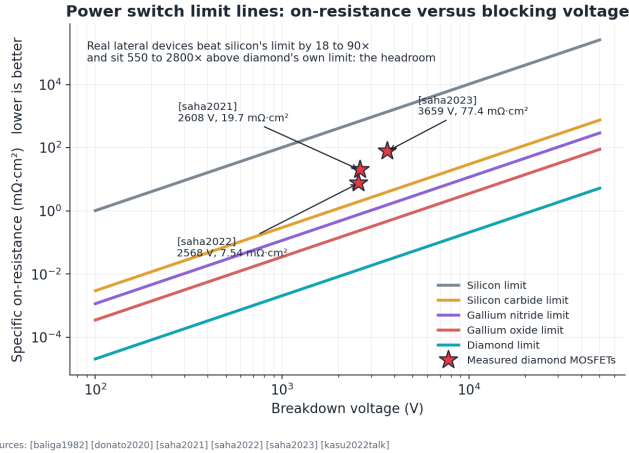


Figure 1: One-dimensional on-resistance limits and measured hydrogen-terminated diamond MOSFETs [6, 7, 8]. Real devices sit above diamond’s own limit but below silicon’s.

3 Process, design kit, and a placed processor

The hydrogen-terminated flow (Fig. 2) creates the channel by surface transfer doping [11], protects it with gold immediately, isolates devices with one oxygen-plasma step, and stabilizes the hole gas with atomic-layer-deposited Al_2O_3 at $\leq 150^\circ\text{C}$ (with NO_2) or $\sim 450^\circ\text{C}$ (for 400°C operation) [12, 13, 14]. Its thermal budget forces growth and the NV anneal (up to 1100°C [15]) ahead of all metal. The traveler gives parameters with sources, pass criteria, safety, and failure modes for each step, and an interactive run log. On this process, the design kit PDK-0 supplies layers, rules, a GDS monitor die, standard cells, LEF abstracts, and a timed Liberty file; the DIA-4 processor synthesizes to 323 cells and places in about 1 mm^2 with pads. An emulator matches its Verilog cycle for cycle on random programs.

4 The two-qubit bottleneck

With the published pair’s parameters, 25 nm spacing, 4.93 kHz single-quantum coupling (a $25\text{ }\mu\text{s}$ double-quantum gate), and echo times of 150 and $514\text{ }\mu\text{s}$ [16], the coherence-limited Bell fidelity is 0.998 . The measured 0.67 corresponds to a preparation probability $q = 0.75$ and the later 0.82 [17] to $q = 0.87$ (Fig. 3). Both match the NV^- fraction under green light [18]. A five-level rate model with measured rates [19] gives a readout contrast of 45% in a 300 ns window and 82% spin polarization; charge-state verification before the gate is therefore the highest-leverage experimental change.

5 Error correction with slow readout

NV readout takes about a millisecond [20, 21], during which data qubits held in nuclear spins idle. Circuit-level simulations with Stim and PyMatching [22, 23] (0.3% link, 1% readout, 0.2% preparation errors) validate a threshold near 1.2% for uniform noise and give an error-suppression factor $\Lambda = 4.03$ at 0.1 ms readout, 3.56 at 1 ms , and 2.64 at 10 ms for a 1 s memory. Distance 7 stops beating distance 3 near 100 ms . Nuclear memories mostly dephase; the standard code wastes this bias because its X memory fails first, while an XZZX code [24, 25], emulated by Hadamard conjugation of a checkerboard sublattice, pushes break-even to about 1000 ms (Fig. 4).

6 How big, how slow

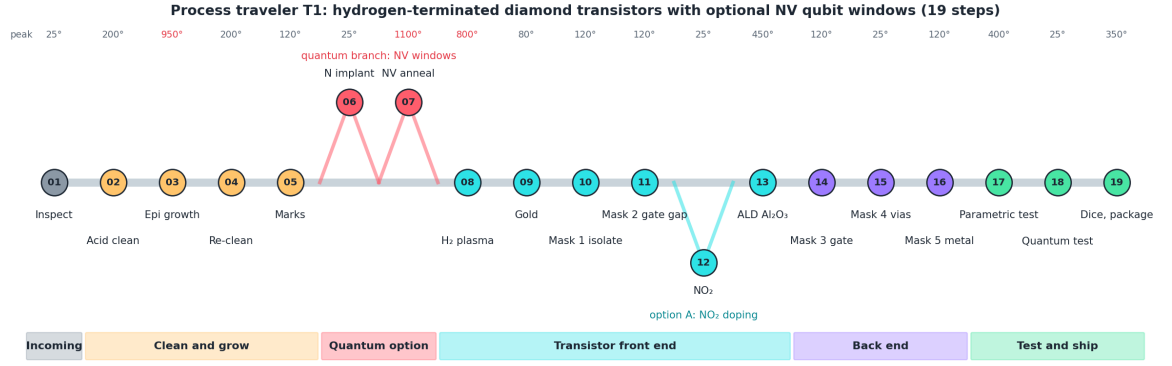
Fitting $p_L(d) = \Lambda\Lambda^{-(d+1)/2}$ [26] and pricing each logical step at d rounds [27] gives, for an RSA-2048-scale job ($6,000$ logical qubits, 3×10^9 steps), distance 57, 78 million physical qubits on a 5.7 mm die, and 6.0 years at 1 ms readout (1.0 year at 0.1 ms), against 4 days for a superconducting machine at its measured $\Lambda = 2.14$ [28] (Fig. 5). The NV figure assumes a 0.3% link error that has not been demonstrated. Once readout is fast, the four dipolar gates dominate each round, so placement precision becomes the next lever.

7 Experiments and teaching

Ten experiments span US\$100 to a pilot line (Fig. 6), following published teaching setups [29, 30, 31]. The cheapest ships with firmware (Raspberry Pi Pico driving an ADF4351 synthesizer) and a host script whose output feeds the same fitting code as a browser Data Lab. A seven-lesson course, 13 interactive labs, and a process traveler turn the framework into teaching material; all browser simulations are tested against the Python models.

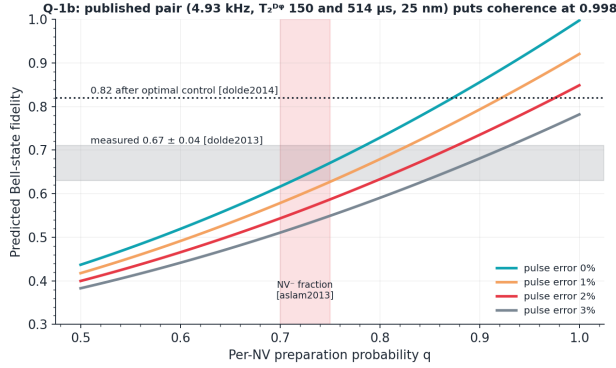
8 Limitations

Transistor and switching parameters are literature-anchored placeholders until the PDK-0 monitor die is measured. The converter model omits gate charge and reverse recovery; the surface-code idle channel is phenomenological at the gate level (depolarizing or pure dephasing), and the XZZX result emulates idle noise without native XZZX hook errors. Workloads in the resource estimate are illustrative scales.



Sources: [kavarada2023] [kasu2012] [pezzagna2010] [hauf2011] [lu2013], adamas.traveler

Figure 2: Process traveler T1: 19 steps from a bare plate to a tested chip, with a quantum branch (NV implant and anneal) and an NO₂ doping option. Heat is spent before the first metal.



Sources: [dolde2013] [dolde2014] [aslami2013], adamas.gate_budget model

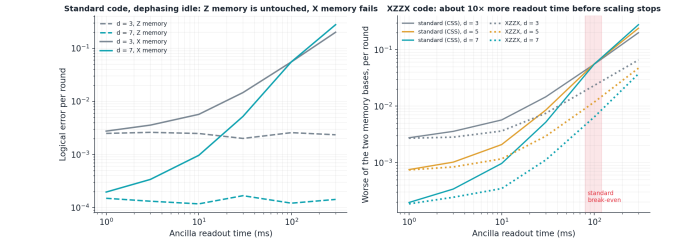
Figure 3: Echoed NV-NV gate budget evaluated with the published pair [16]: fidelity versus charge-state preparation probability. Coherence limits fidelity to 0.998.

Reproducibility

`pip install -e ".[dev]" && python -m pytest -q && make figures paper` regenerates every figure, the numbers in this text (`python -m adamas.paper`), and this PDF. Interactive version: normansrule.github.io/adamas-diamond-framework.

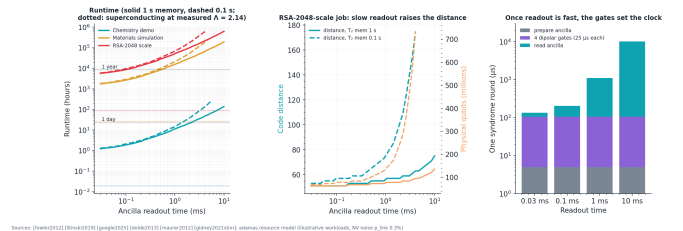
References

- [1] B. J. Baliga. Semiconductors for high-voltage, vertical channel field-effect transistors. *Journal of Applied Physics*, 53:1759, 1982.
- [2] L. Wei, P. K. Kuo, R. L. Thomas, T. R. Anthony, and W. F. Banholzer. Thermal conductivity of isotopically modified single crystal diamond. *Physical Review Letters*, 70:3764, 1993.
- [3] J.-P. Lagrange, A. Deneuville, and E. Gheeraert. Activation energy in low compensated homoepitaxial boron-doped diamond films. *Diamond and Related Materials*, 7:1390, 1998.



Sources: [buckley2018] [benham2021] [gibney2021] [hoggott2021] [maure2021], adamas.circuit_gen: QXXZ emulated by Hadamard conjugation of a checkerboard subcircuit

Figure 4: Circuit-level surface code with dephasing idle errors during readout. Left: standard code, Z and X memories. Right: worse basis for standard and XXXX codes.



Sources: [maure2021] [benham2021] [gibney2021] [hoggott2021] [maure2021], adamas.resource_model: [adamashomebook], NV read, 3, 3, 3, 3, 3

Figure 5: Resource estimate from the circuit-level fits: runtime, code distance, and the composition of one syndrome round.

- [4] Semiconductor Today. Element Six and Orbray accelerate wafer-scale single-crystal diamond for volume production. Semiconductor Today (news), 2026. 16 June 2026.
- [5] M. Liao, H. Sun, and S. Koizumi. High-temperature and high-electron mobility metal-oxide-semiconductor field-effect transistors based on n-type diamond. *Advanced Science*, 11(13):2306013, 2024.
- [6] N. C. Saha, S.-W. Kim, T. Oishi, Y. Kawamata, K. Koyama, and M. Kasu. 345-MW/cm² 2608-V NO₂ p-type doped diamond MOSFETs with an Al₂O₃ passivation overlayer on heteroepitaxial diamond. *IEEE Electron Device Letters*, 42:903, 2021.
- [7] N. C. Saha, S.-W. Kim, T. Oishi, and M. Kasu. 875-MW/cm² low-resistance NO₂ p-type doped chemical me-

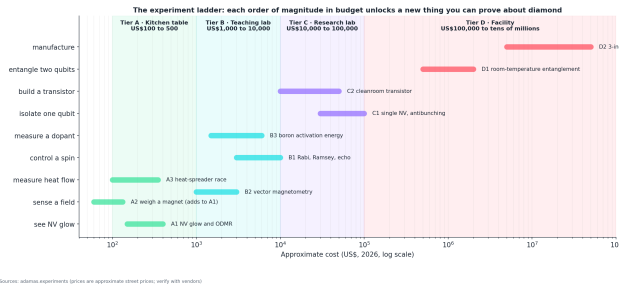


Figure 6: Ten experiments in four budgets, with parts, steps, safety, and simulated expected results; a firmware kit makes the US\$150 to 400 ODMR experiment runnable.

chanical planarized diamond MOSFETs. *IEEE Electron Device Letters*, 43:777, 2022.

- [8] N. C. Saha, S.-W. Kim, K. Koyama, T. Oishi, and M. Kasu. 3659-V NO₂ p-type doped diamond MOSFETs on misoriented heteroepitaxial diamond substrates. *IEEE Electron Device Letters*, 44:112, 2023.
- [9] R. W. Erickson and D. Maksimovic. *Fundamentals of Power Electronics (3rd ed.)*. Springer, 2020.
- [10] J. Pernot, P. N. Volpe, F. Omnes, et al. Hall hole mobility in boron-doped homoepitaxial diamond. *Physical Review B*, 81:205203, 2010.
- [11] F. Maier, M. Riedel, B. Mantel, J. Ristein, and L. Ley. Origin of surface conductivity in diamond. *Physical Review Letters*, 85:3472, 2000.
- [12] M. Kasu, H. Sato, and K. Hirama. Thermal stabilization of hole channel on H-terminated diamond surface by using atomic-layer-deposited Al₂O₃ overlayer and its electric properties. *Applied Physics Express*, 5:025701, 2012.
- [13] H. Kwarada, H. Tsuboi, T. Naruo, et al. C-H surface diamond field effect transistors for high temperature (400 C) and high voltage (500 V) operation. *Applied Physics Letters*, 105:013510, 2014.
- [14] H. Kwarada. Diamond p-FETs using two-dimensional hole gas for high frequency and high voltage complementary circuits. *Journal of Physics D: Applied Physics*, 56:053001, 2023.
- [15] S. Pezzagna, B. Naydenov, F. Jelezko, J. Wrachtrup, and J. Meijer. Creation efficiency of nitrogen-vacancy centres in diamond. *New Journal of Physics*, 12:065017, 2010.
- [16] F. Dolde, I. Jakobi, B. Naydenov, et al. Room-temperature entanglement between single defect spins in diamond. *Nature Physics*, 9:139, 2013.
- [17] F. Dolde, V. Bergholm, Y. Wang, et al. High-fidelity spin entanglement using optimal control. *Nature Communications*, 5:3371, 2014.
- [18] N. Aslam, G. Waldherr, P. Neumann, F. Jelezko, and J. Wrachtrup. Photo-induced ionization dynamics of the nitrogen vacancy defect in diamond investigated by single-shot charge state detection. *New Journal of Physics*, 15:013064, 2013.
- [19] J.-P. Tetienne, L. Rondin, P. Spinicelli, et al. Magnetic-field-dependent photodynamics of single NV defects in diamond: an application to qualitative all-optical magnetic imaging. *New Journal of Physics*, 14:103033, 2012.
- [20] P. Neumann, J. Beck, M. Steiner, et al. Single-shot readout of a single nuclear spin. *Science*, 329:542, 2010.
- [21] D. A. Hopper, H. J. Shulevitz, and L. C. Bassett. Spin readout techniques of the nitrogen-vacancy center in diamond. *Micromachines*, 9:437, 2018.
- [22] C. Gidney. Stim: a fast stabilizer circuit simulator. *Quantum*, 5:497, 2021.
- [23] O. Higgott. PyMatching: a Python package for decoding quantum codes with minimum-weight perfect matching. *ACM Transactions on Quantum Computing*, 3:16, 2022.
- [24] D. K. Tuckett, S. D. Bartlett, and S. T. Flammia. Ultra-high error threshold for surface codes with biased noise. *Physical Review Letters*, 120:050505, 2018.
- [25] J. P. Bonilla Ataides, D. K. Tuckett, S. D. Bartlett, S. T. Flammia, and B. J. Brown. The XZZX surface code. *Nature Communications*, 12:2172, 2021.
- [26] A. G. Fowler, M. Mariantoni, J. M. Martinis, and A. N. Cleland. Surface codes: towards practical large-scale quantum computation. *Physical Review A*, 86:032324, 2012.
- [27] D. Litinski. A game of surface codes: large-scale quantum computing with lattice surgery. *Quantum*, 3:128, 2019.
- [28] Google Quantum AI and Collaborators. Quantum error correction below the surface code threshold. *Nature*, 638:920, 2025.
- [29] J. Stegemann et al. Modular low-cost 3D printed setup for experiments with NV centers in diamond. *European Journal of Physics*, 44:035402, 2023.
- [30] H. Williams, A. Newman, S. Graham, C. Stephen, and G. Morley. A portable LED-based diamond magnetometer for outreach and teaching labs. *American Journal of Physics*, 94:489, 2026.
- [31] V. K. Sewani, H. H. Vallabhapurapu, Y. Yang, et al. Coherent control of NV- centers in diamond in a quantum teaching lab. *American Journal of Physics*, 88:1156, 2020.